Thunderbolt 3 - One port to rule them all

The soon to be ubiquitous interface deserves your attention

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When Thunderbolt first hit the scene back in 2011 and only Apple decided to implement the same. It was rumoured to be a Mac only interface but Intel soon put those rumours to rest. One thing was obvious, the industry didn't seem to bother with the new interface. Not many devices were ready to move away from USB and adopt Thunderbolt. For a time it seemed that Thunderbolt would end up as the next FireWire – the last guy to fall in battle against the USB.

The aim was clear from the beginning, to create an interface that would serve more than one purpose. It would provide video data and raw data transfer, so it was only natural for the port to be chosen from the existing array of display interface. The first Thunderbolt was shaped in the form of the miniDisplayPort and thus, you could connect a display to it if required or hook up a storage drive. The DP was rigid and offered enough bandwidth – 10 Gbps – for high-throughput data transfers. But it was way ahead of its time. Mainstream desktop motherboards wouldn't dare add the extra Thunderbolt chip and lose out on market share since the extra chip would simply make things a little expensive (read up on licensing agreements to know why). At most, motherboard manufacturers would slip in a Thunderbolt AIC (Add-in-Card) header so that the user, if they ever needed could buy a Thunderbolt AIC and slap it on their boards. Needless to say, this approach of meeting things half-way didn't catch on and Thunderbolt 2 went the same way as Thunderbolt 1 despite showing promise. ASUS was the first motherboard manufacturer to adopt Thunderbolt 2.

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Thunderbolt 2 had double the bandwidth of the previous generation but it required a lot of power which meant it couldn't be adopted onto portable devices. It allowed streaming 4K video back when 4K monitors were rarely available unless you had really deep pockets. So again, the port was way ahead of its time which prevented wide-scale adoption.

Which brings us to Thunderbolt 3. The time was now ripe, thanks to the advent of SSDs, existing data transfer protocols were already saturated and multiple new interfaces started popping up. For internal devices, there's SATA Express and M.2 (NGFF) and while the battle is yet to be won, it seems that M.2 will win the crown. For portable devices, USB 3.0 arrived and it was soon followed by USB 3.1 heralding the new USB Type-C connector. Since USB ended up being quickly adopted, the same was bound to happen with USB Type-C, so the choice of interface for the third iteration of Thunderbolt was glaringly obvious – USB Type-C. Thunderbolt 3 can handle 40 Gbps which is enough to drive two 4K displays or one 5K display and with the ability to channel 100 Watts, more power hungry peripherals can be made portable. The average laptop charger channels less than 100 watts and 40 Gbps is enough to transfer a full-length 4K feature film in under 30 seconds.

And thus, thanks to USB Type-C, the Thunderbolt interface will now become truly popular. Every major manufacturer is adopting the USB Type-C connector, we started seeing it on not just premium devices but even extremely budget friendly devices like Mini-PCs that barely cost ₹9,500 rupees. Plenty of mobile phones have adopted the Type-C connector and so have laptops. In fact, between CES and Computex, we've seen Type-C being slapped onto everything under the sun. In the last month, we've even seen external graphics solutions like the Razer Core which uses a Thunderbolt 3 connector. While peripherals may be a little expensive at the moment, the rapid adoption will soon bring down costs. Thunderbolt 3 can be used to daisy 6 devices and you can even have a 4K display being the 6th device. So with Thunderbolt adopting USB Type-C as the port of choice, this could very well be the year Intel was looking forward to. Versatility is the name of the game and Thunderbolt 3 seems to have checked all the right boxes. d